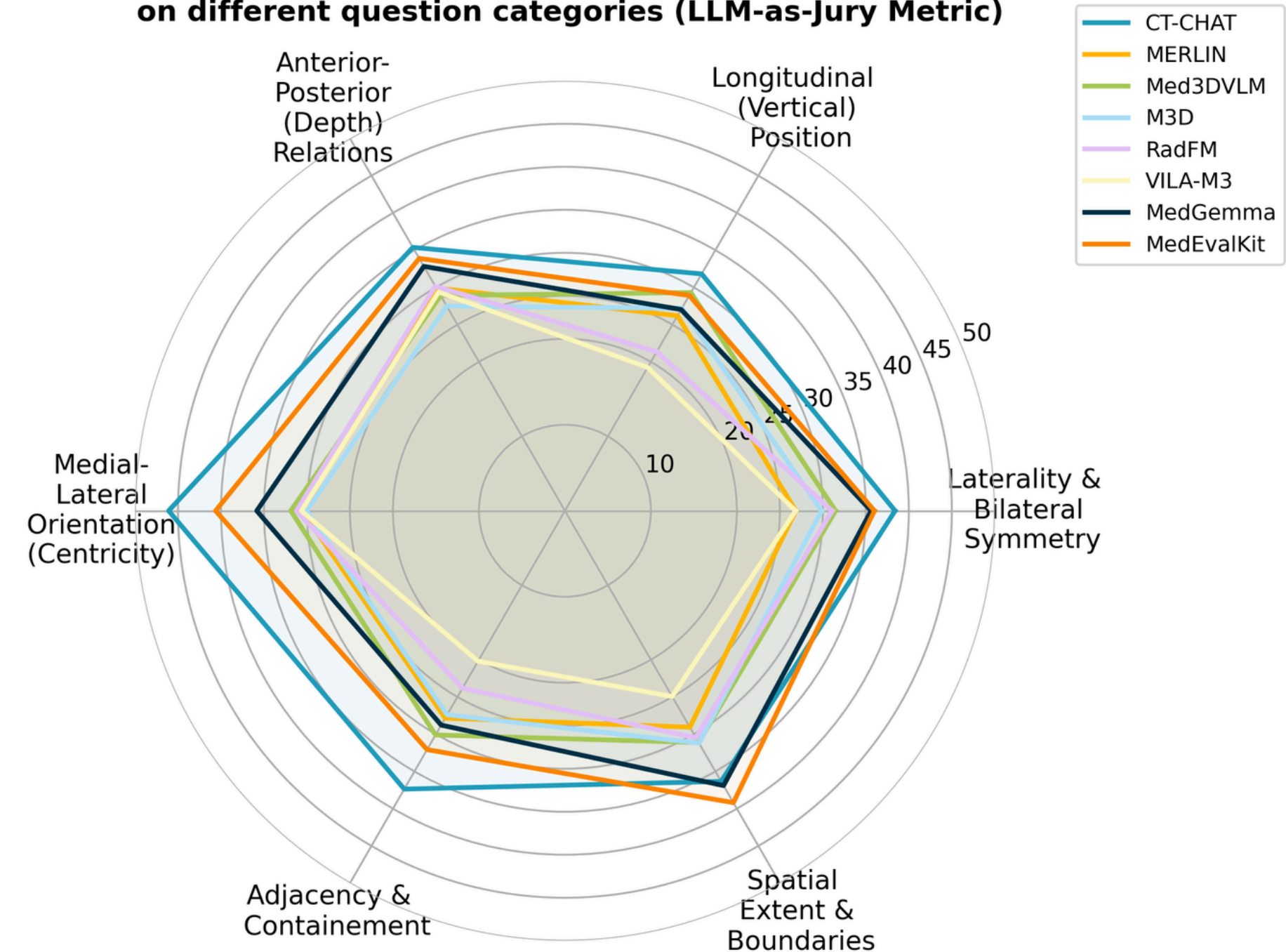


Performance of Medical 3D VLMs on the CT-SpatialVQA Benchmark
on different question categories (LLM-as-Jury Metric)



Category	Spatial Relevance
• Laterality & Bilateral Symmetry	• Consistent left-right anatomical grounding relative to the sagittal midline; prevents laterality inversion and side hallucination.
• Longitudinal (Vertical) Position	• Requires volumetric integration along the superior-inferior axis and enforces slice-level consistency across the 3D stack.
• Anterior-Posterior (Depth) Relations	• Preserves depth cues and front-back anatomical arrangement essential for compartment-level reasoning.
• Medial-Lateral Orientation (Centricity)	• Maintains stable central-peripheral localization and global anatomical reference frames.
• Adjacency & Containment	• Encodes topological relations, distinguishing touching from containment within anatomical boundaries.
• Spatial Extent & Boundaries	• Tracks spatial continuity, confinement, and compartment crossing to model disease spread accurately.